Muram Yaser Basher

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ACADEMIC PROFILE

Final-year Software Engineering undergraduate interested in artificial intelligence, computer vision, machine learning, human-computer interaction, and accessible educational technologies. Experience developing AI-enabled prototypes, full-stack web applications, and embedded assistive systems.

EDUCATION

INES-Ruhengeri, Rwanda | Bachelor's degree in Software Engineering | Expected graduation: 2027

Reported cumulative average: 4.0/5.0.

RESEARCH INTERESTS

Computer vision and visual perception; machine learning; multimodal and assistive AI; real-time human-computer interaction; efficient AI for resource-constrained environments; technology-enhanced learning.

SELECTED TECHNICAL AND RESEARCH-ORIENTED PROJECTS

VisionBridge — AI-Powered Assistive Smart Glasses | Team project

- Helped develop a wearable assistive prototype integrating object/obstacle detection, text reading, and spoken feedback to support people with visual impairments.
- Worked with computer vision and embedded computing components, including Raspberry Pi hardware, a camera, distance sensing, and audio output.
- Project recognized with second place at Hult Prize Rwanda nationals (team award).

Interactive Computer Vision-Based Projection System | Independent final-year project prototype | 2026–present

- Developing a system that combines camera-based hand-gesture recognition with projected interactive content on ordinary surfaces.
- Implemented and tested fingertip detection and gesture-controlled object dragging using a camera prototype; exploring applications in accessible, multilingual learning.

Opportune — AI-Powered Opportunity Tracking Platform | Independent software project | 2026

- Built a full-stack web application to extract structured details from opportunity links or pasted text, track application progress, and support deadline management.
- Implemented a Django REST Framework backend, React/Vite interface, JWT-based authentication, AI-assisted extraction, and email reminder functionality.
- Advanced through the first round of the Code for Impact 2026 competition with this project.

Machine Learning for Agricultural Yield Prediction | Academic coursework | 2026

- Built preprocessing and regression pipelines using Python and scikit-learn; compared baseline, linear regression, and random forest models on agricultural data.
- Selected linear regression based on evaluation performance; explored classification, clustering, and API-based predictions.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Sudanese Students Association, INES-Ruhengeri | Media Secretary; later Head of Media

- Helped establish and grow communications for the university's Sudanese student community, supporting students adapting to study and life in Rwanda.
- Contributed to content planning and digital engagement across Instagram, WhatsApp, and TikTok, alongside a student leadership team.

Sudanese Students English Club | Founder

- Initiated an English-learning community to help newly arrived students strengthen communication skills and participate more confidently in university life.

AWARDS AND ADDITIONAL DEVELOPMENT

- Second Place — Hult Prize Rwanda National Competition, VisionBridge team.
- Best Female Speaker — INES Debate Club competition, 2026.
- Participated in a one-week training program at Carnegie Mellon University Africa related to the [the Bridge programe].

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, PHP. Frameworks/tools: Django, Django REST Framework, React, Vite, FastAPI, Expo/React Native, Git/GitHub. AI/data: scikit-learn, OpenCV, TensorFlow Lite, Edge Impulse, data preprocessing, model evaluation. Hardware: Raspberry Pi, camera and sensor integration. Databases: PostgreSQL, MySQL.

LANGUAGES AND TESTING

Arabic and English; Duolingo English Test: 116 (2026).